Design and Implementation of a Priority-Based Music Scheduling System Using Queues and Binary Trees

Abstract
This project presents the design and implementation of a priority-based music scheduling system inspired by the television program *Night Ahangi*. The system processes user music requests, categorizes them using queues, and selects songs for broadcasting based on priority using a binary tree structure implemented as a priority queue. The goal is to maximize audience satisfaction within a fixed program duration of one hour.

Problem Statement
Music requests are received from users in four categories: Traditional (S), Pop (P), Classical (K), and Local (M). Each request includes the music category, number of requests, and duration of the song. The system must determine which songs are broadcast during a one-hour program while respecting time constraints and prioritizing songs with higher demand.

Methodology
The system first stores all incoming music requests in an input queue. Requests are then separated into category-specific queues. A priority queue (binary max-heap) is used to select songs with the highest number of requests. Songs are played until the total broadcast time reaches the one-hour limit.

Data Structures Used
- Queue: for managing input and category-based music requests.
- Priority Queue (Binary Tree): for selecting the next song based on maximum user demand.
- Classes and Objects: for modeling songs and scheduling logic in an object-oriented manner.

Results and Output
The program outputs a detailed report showing the total playtime, satisfaction level, and percentage of broadcast time allocated to each music category. This demonstrates efficient scheduling and fair utilization of available broadcast time.

Conclusion
This project demonstrates the practical application of queues and binary trees in solving real-world scheduling problems. The modular and object-oriented design improves readability, maintainability, and scalability of the system.